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Comments on “E-Voting Meets Blockchain: A Survey”

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Name	Type of File	Size	Page
comment_13_8_2026.pdf	Main Document - PDF	235.0 KB	Page 3

Comments on “E-Voting Meets Blockchain: A Survey”

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Abstract—Due to the enormous number of papers discussing variant e-voting systems, researchers sometimes mix the facts between what is proposed and what is currently existing, which may lead to a chain of misconceptions by other researchers reading and/or citing their paper. This comment aims to correct some mixed facts in a review article, about the use of blockchains in e-voting systems; the article [1] appeared in 2023, vol (11), and is cited by 87 papers or more. The article claims in its abstract and in section IV.A that countries like Germany, Estonia, Australia and Switzerland have deployed blockchain technology in their e-voting systems, which is not true; although those countries may have deployed blockchains in other governmental aspects and/or their private sectors may hold blockchain-based e-voting systems. This comment does not aim to contradict the main concepts included in the paper, since there are other countries like Russia, South Korea and UAE that make use of the blockchain technology in their e-voting systems; however, the milestone MIT paper, [2], should have been referenced when discussing the risks and challenges.

Keywords—e-voting, blockchains, IVXV, Swiss-Post

I. PROBLEMS/ISSUES IN THE PAPER

The paper in [1] (2023) surveys a very rich topic in the literature which is online voting systems that uses blockchain technology; existing systems and/or research vary from small sized private entities to large scale nationwide elections. However, there are some issues that need to be revised/handled/corrected.

A. The survey should include papers with opposing point of view

When largescale political elections is reviewed in the survey (Table6 and A in section IV), this should definitely include a credible debatable reference as [2] to be just skipped; a paper whom authors include Ron Rivest, Bruce Schneier and most known pioneers in cryptography research. Even if the authors believe [2] presents an old school paradigm that is suspicious against all kinds of internet voting in political elections, it is basic that papers with an opposing point of view should be included, and discussed, in review articles.

B. Wrong countries in the abstract

Then comes the most important correction regarding the mentioned countries; none of the countries mentioned in the abstract of [1] uses blockchains in governmental online voting. Germany does not allow online voting at all in nation level elections (although sometimes used by private entities and political parties), while Estonia, Russia and Switzerland do not deploy blockchain technology in their internet voting systems.

1) Germany

According to [3,4], the constitutional German court rejected electronic voting since 2019 because they considered cryptographic proofs “somewhat untenable as a verification”

as they require expert testimony; the verdict itself can be found in [5]. The reference cited in the paper as “[73]” (polyas.com) is clearly a commercial one for the company that does work in Germany, but for the private sector; this paper [6] for example discusses the German Informatics Society (GI) elections using Polyas. In addition, the company stated clearly since 2021, [7], that it preferred centrally stored Public Bulletin Boards over decentralized blockchains ledgers. Hence, the lines about Germany in page 23301 of the manuscript are wrong and should be corrected.

2) Estonia

Estonia only uses a private permissioned blockchain (KSI) in its e-government [8,9], but it does not use blockchain technology anywhere in the Estonian internet voting system (IVXV); the references “[67,68]” in the paper are about KSI and the e-government, not e-voting. In fact, this Estonian research article [10] in 2018 is clear evidence, where it discusses why did Estonia rejected many submitted proposals for deploying blockchain in their internet voting system; although could not access the full pdf, we found also a thesis from Tartu university, [11] in 2019, that discusses the subject. If those references seems old, the interested reader can check the current architecture [12] design and the GitHub site [13] for the code.

3) Switzerland Swiss-Post

The Luxoft municipal elections referred to in the paper in subject, [14,15], was only a trial one that was conducted in the city of Zug in 2018 using 72 citizens out of 220 holding digital identities; the trial vote was on things like whether fireworks should be allowed on festivals or not. However, Swiss Post design, the current online voting system in Switzerland [16] does not use blockchains. The Luxoft experiment is already mentioned again in the 2nd column of the same page 23302 under “B. Commercial Systems...”

4) Australia

Australia is like Germany in not deploying internet voting in nationwide elections. However, it allows commercial use and had some pilot studies. The reference cited in the paper as “[72]” does not exist, maybe changed through the years, the correct site is (<https://bitcoin.com.au/south-australian-government-blockchain-technology-elections/>)

5) Norway

The facts stated in the paper about Norway stopping e-voting is correct. Ironically, given all the above, the paper does not mention their blockchain-based online voting experiment, TIVI [17,18], in 2018; this happened at the country level, during a localized referendum, and was offered by Smartmatic company (since the paper is also listing the

working companies in Tables 6,7). It is also worth mentioning that this company is partnering with Cybernetica for designing IVXV in Estonia, and this pilot was conducted in the same period as [10,11].

II. SUMMARY & CONCLUSION

The paper in subject contains some **mixed-up facts about countries that deploy blockchains in their governmental e-voting systems**. Estonia, Switzerland e-voting systems do not use blockchains; while Germany and Australia do not allow e-voting at all in nationwide elections although it is widely used by the private sector, but not necessarily with blockchains. The paper mixes pilot studies with current working systems (since 2023); pilot studies should have been moved up in the “Academic proposals” section III. This would have given a chance to discuss or at least mention the design choices of each trial case; for example, the blockchain choice since the paper has dedicated complete subsections in II for different blockchain kinds and consensus protocols. The contents of section II is beyond the scope of this comment which was mainly incentivized by the wrong facts in the abstract. These facts **need to be corrected**; also, **papers presenting an opposing point of view** (ex.: [2,10,19]) **should be included in the survey**.

It is concerning that those mistakes remained unnoticed from March 2023 till Aug 2026 with 87 citations recorded on the IEEE site and 135 on Researchgate¹; this is why the author included many references to prove each fact. This Vietnamese legal journal² paper, [20], which references the paper in subject as “[35]” is an example of the *chain of misconception* mentioned in the abstract. The line in the abstract must be corrected and the last two statements in the first paragraph of the 2nd column of page 23301 too; however, it might be practically hard to perform more than minor changes to the manuscript. Removing those countries with their associated references would cause a lot of mess. Moving them in a separate section/ subsection would correct the facts in an easy neat way; it could be dedicated to countries that rejected the use of blockchains in their e-voting systems and may also contain a couple of lines about papers against blockchain usage in e-voting.

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¹ https://www.researchgate.net/publication/369044354_E-Voting_Meets_Blockchain_A_Survey/citations

² E-voting papers are sometimes cited by legal/political researchers who are not experts in the technicalities and consider peer-reviewed articles as a credible source of information